

IDS responses can be categorised into two main types:

- **Active defence:** An active defence is ‘real-time’ defence where the system initiates an action at the moment of detection rather than waiting for a report from another system.
- **Passive defence:** In this type of defence, IDS only responds after receiving and processing information that an intrusion attempt has taken place.

The goal of an AI and ML based algorithm is to optimise certain features and improve its classifiers, so that it reduces the number of false alarms that come up while trying to identify an intruder.

**Phishing detection system:** An AI based system detects phishing emails by analysing the content of the email and comparing it with a database of known phishing emails. The phishing detection system (PDS) can also detect if the sender is spoofing another person’s identity.

PDS can also be used with voice, video, and image messages. The system activates when users receive a suspicious email or when they send an email containing personal information. Some of the features of the PDS are:

- Automatically detects email phishing scams
- Stores emails with malicious content in a quarantine folder
- Triggers user notifications when the system detects a new virus in an email

- Maintains detailed logs of all email activity
- Detects emails that contain phishing links
- Automatically generates a report of every detected email

The goal of the PDS is to automatically detect and report emails that contain phishing links.

**Automated processes that optimise human analysis:** Automated processes can be set up by analysing reports on past actions generated by security analysts to identify and respond to certain attacks successfully. AI algorithms use this knowledge to build a model, which can be used later for identifying similar cyber activities. Using this model, AI algorithms

| Feature                              | Open source tool/product              | Remarks                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Malware detection and identification | REMnux, OpenEDR                       | <b>REMnux</b> is a free Linux toolkit for reverse engineering and analysing malware. <b>OpenEDR</b> helps organisations to secure their infrastructure against malware, ransomware, data breaches and other threats.                                                                                                                                                                          |
| Threat analysis                      | Nmap, Metasploit                      | <b>Nmap</b> provides methods to find open ports, detect host devices, verify active network services, fingerprint operating systems and locate potential backdoors. <b>Metasploit</b> helps security professionals perform simulation attacks to find loopholes in a system.                                                                                                                  |
| Spam detection                       | FortiClient                           | <b>FortiClient</b> reduces the risk of malware, and blocks spam URLs as well as exploit kits.                                                                                                                                                                                                                                                                                                 |
| Network intrusion detection          | Security Onion, Snort, PfSense, OSSEC | <b>Security Onion</b> provides network monitoring via full packet capture, host-based and network-based intrusion detection systems, log indexing, and search and data visualisation features. <b>Snort</b> is capable of real-time traffic analysis and logging. <b>PfSense</b> is configured for intrusion detection and prevention, traffic shaping, load balancing and content filtering. |
| Phishing detection                   | Gophish                               | <b>Gophish</b> provides a full-featured toolkit for security administrators to build their own phishing campaigns.                                                                                                                                                                                                                                                                            |
| Asset discovery                      | AlienVault, OSSIM                     | <b>OSSIM</b> offers end-to-end security information and event management through asset discovery, behavioural monitoring, and event correlation.                                                                                                                                                                                                                                              |
| Incident response                    | OpenVAS, Nikto                        | <b>OpenVAS</b> is an all-in-one vulnerability scanner. It tests for security issues, mis-configured systems and outdated software.                                                                                                                                                                                                                                                            |
| SQL injection flaw detection         | Sqlmap                                | <b>Sqlmap</b> automates detecting and exploiting SQL injection flaws of database servers, enabling a remote hacker to take control.                                                                                                                                                                                                                                                           |

Table 1: Popular open source AI and ML based cyber security tools